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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS HACKS STANDINGS CUSTOM INVOCATION

F1. Guess the K-th Zero (Easy version)

time limit per test: 1 second
memory limit per test: 256 megabytes

This is an interactive problem.

This is an easy version of the problem. The difference from the hard version is that in the easy version $t = 1$ and the number of queries is limited to 20.

Polycarp is playing a computer game. In this game, an array consisting of zeros and ones is hidden. Polycarp wins if he guesses the position of the k -th zero from the left t times.

Polycarp can make no more than 20 requests of the following type:

- $? l r$ — find out the sum of all elements in positions from l to r ($1 \leq l \leq r \leq n$) inclusive.

In this (easy version) of the problem, this paragraph doesn't really make sense since $t = 1$ always. To make the game more interesting, each guessed zero turns into one and the game continues on the changed array. More formally, if the position of the k -th zero was x , then after Polycarp guesses this position, the x -th element of the array will be replaced from 0 to 1. Of course, this feature affects something only for $t > 1$.

Help Polycarp win the game.

Interaction

First, your program must read two integers n and t ($1 \leq n \leq 2 \cdot 10^5, t = 1$).

Then t lines follow, each of which contains one integer k ($1 \leq k \leq n$). It is guaranteed that at the moment of the request the array contains at least k zeros. In order to get the next value of k , you must output the answer for the current value of k .

After that, you can make no more than 20 requests.

Use the following format to output the answer (it is not a request, it doesn't count in 20):

- $! x$ — position of the k -th zero.

Positions in the array are numbered from left to right from 1 to n inclusive.

After printing t answers, your program should exit immediately.

In this task, the interactor is **not adaptive**. This means that within the same test, the hidden array and the queries **do not change**.

In case of an incorrect query, -1 will be displayed. When this value is received, your program must immediately exit normally (for example, by calling `exit(0)`), otherwise, the testing system may issue an arbitrary verdict.

If the number of requests is exceeded, the verdict *wrong answer* will be displayed.

Your solution may get the verdict *Idleness limit exceeded* if you don't print anything or forget to flush the output buffer.

To flush the output buffer, you need to do the following immediately after the query output and the end-of-line character:

- `fflush(stdout)` or `cout.flush()` in C++;
- `System.out.flush()` in Java;
- `flush(output)` in Pascal;
- `stdout.flush()` in Python;
- see the documentation for other languages.

Hacks

Use the following format for hacks:

On the first line print the string s ($1 \leq |s| \leq 2 \cdot 10^5$), consisting of zeros and ones, and an integer t ($t = 1$) — hidden array and number of requests, respectively. In the next t lines output the number k ($1 \leq k \leq |s|$).

The hacked solution will not have direct access to the hidden array.

Codeforces Round 719 (Div. 3)

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
361335502	Feb/03/2026 21:32	Accepted
361334910	Feb/03/2026 21:28	Wrong answer on test 17
361333434	Feb/03/2026 21:19	Wrong answer on test 4

→ Problem tags

binary search interactive *1600

No tag edit access

→ Contest materials

- Announcement (en) ✕
- Editorial (en) ✕

Example

input

Copy

6 1
2

2

1

1

0

0

output

Copy

? 4 6

? 1 1

? 1 2

? 2 2

? 5 5

! 5

Note
In the first test, the [1, 0, 1, 1, 0, 1] array is hidden. In this test $k = 2$.

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